

Machine Learning Wireless Communications - Assignment 2

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GitHub repository: https://github.com/Kandi-Rohan/MLWC_2

This report presents the required experiments, figures, and observations. The supplied channel generator, parameters, and random seed were preserved when regenerating the LOS/NLOS dataset.

Question 1: SVM for LOS/NLOS Identification

Part (b): Classification at Each SNR

Method. For every SNR value, six RBF SVM classifiers with $C=1$ were trained using an 80/20 split. StandardScaler was fitted only on the training portion and then applied to the test portion. SVM-1 through SVM-5 use one feature each; SVM-6 uses all five features.

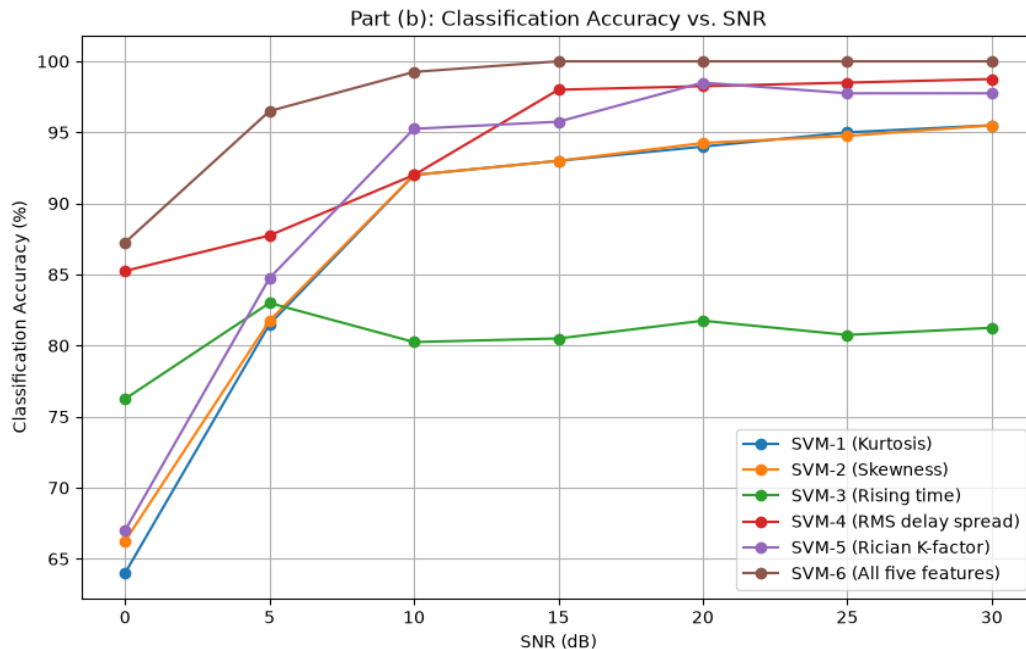


Figure 1 observation. RMS delay spread is the strongest single feature overall, reaching about 98% at high SNR. It captures the physical difference between the short LOS delay profile and the more dispersed NLOS profile. The corrected Rician K-factor also becomes highly discriminative as noise decreases, because the LOS dominant path produces a strong peak-to-rest power ratio.

Feature combination. SVM-6 reaches 87.25% at 0 dB. Its largest improvement is at 5 dB, where it is 8.75 percentage points above the best single feature. The gap is about 4.0 points at 10 dB and about 1.25-2.0 points from 15-30 dB, showing the value of combining complementary channel features.

Lowest-SNR feature. Kurtosis is the lowest-performing feature at the very lowest SNR values in this experiment, approximately 62.75% at 0 dB and 82.25% at 5 dB. Because it is a fourth-order statistic of received power, additive noise strongly changes the estimated tails and makes the statistic unreliable. Rising time is also noise-sensitive because noise can change the apparent strongest tap, but its curve is not the lowest at 0-5 dB.

Part (c): Fixed-SNR Training

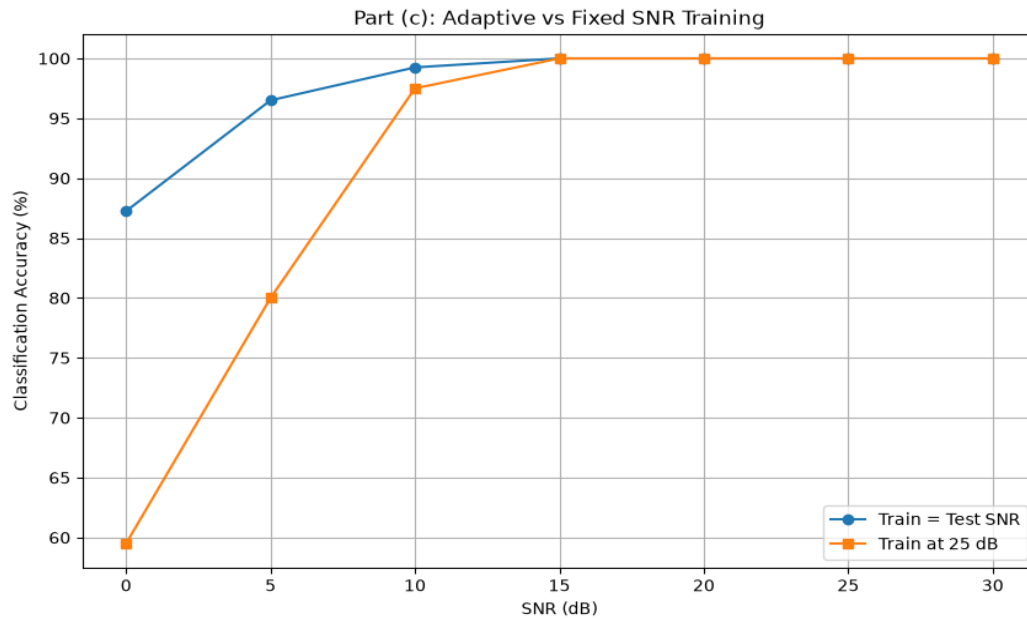


Figure 2 observation. The matched-SNR model generally performs better at low SNR because it learns feature distributions that include the distortion at that operating condition. The model trained only at 25 dB is trained on cleaner features and therefore loses accuracy when applied to heavily noisy 0-10 dB data. At high SNR, the curves become close because the feature distributions stabilize as the noise becomes small.

Low-SNR interpretation. Training at 0 dB and testing at high SNR would also create a distribution mismatch: the model would learn boundaries influenced by severe noise and then apply them to tightly clustered, cleaner features. Training across representative SNR values would be a practical way to improve generalization.

Question 2: 16-QAM Demodulation and K-means

Part (a): Dataset and Features

A square 16-QAM constellation with I,Q levels $\{-3,-1,1,3\}$ was normalized to unit average symbol energy. Two hundred noisy samples were generated per symbol at each SNR using the fixed seed 67. For each received sample, instantaneous amplitude r and phase θ were extracted in addition to the Cartesian coordinates.

Part (b): Choosing K and Comparing Feature Sets

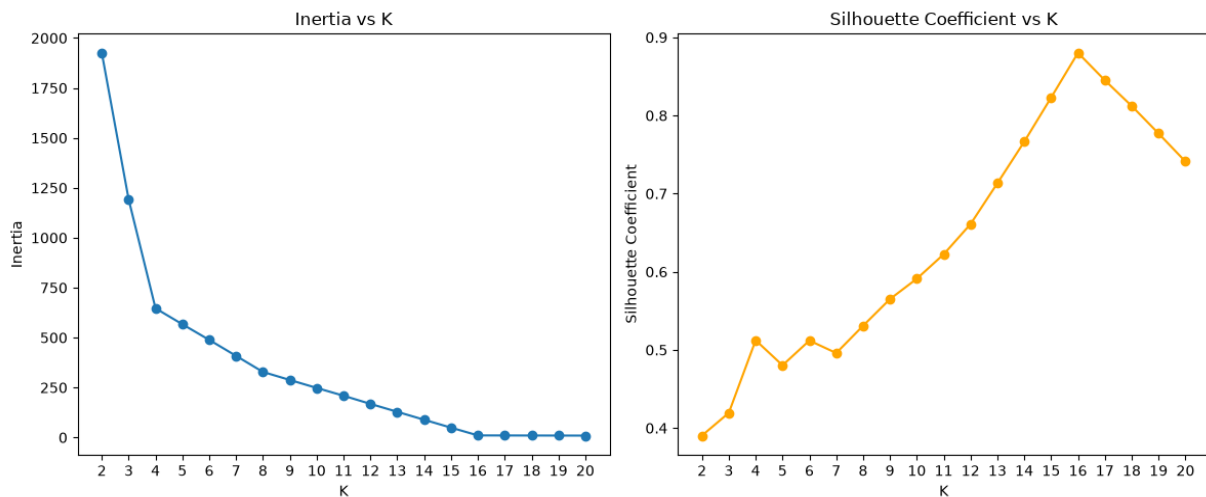


Figure 3 observation. Inertia decreases as K increases, with the main reduction occurring before the true 16-symbol structure is represented. The silhouette curve provides a cluster-separation check. $K=16$ is the appropriate choice because it is supported by the diagnostics and by the known number of constellation points.

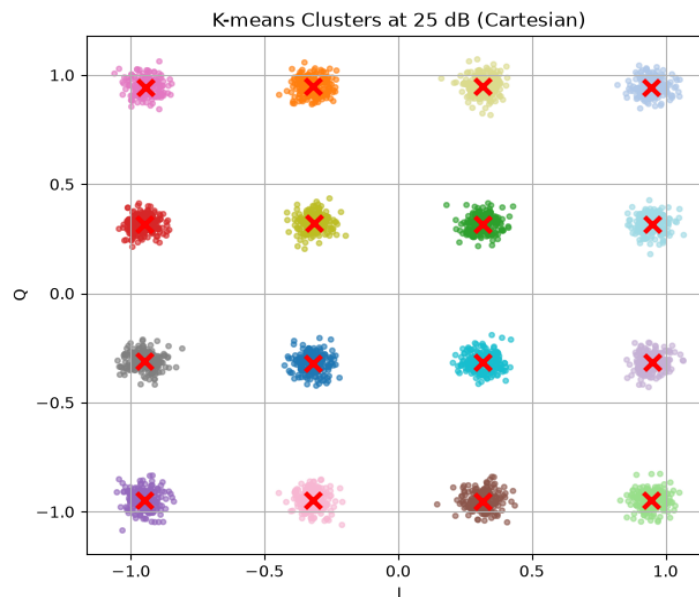


Figure 4 observation. At 25 dB, the received samples form 16 compact groups around the QAM grid and the learned centroids lie near those groups. Cartesian Euclidean distance matches the physical geometry of square QAM, so the Voronoi regions have the expected grid-like interpretation.

Cluster purity results at 25 dB.

Feature set	Purity
Cartesian (r_x_I , r_x_Q)	1.0000
Polar (r , θ)	0.5172
Combined, standardized	1.0000

Geometry observation. Cartesian coordinates preserve the actual QAM grid spacing. In unweighted polar coordinates, equal changes in angle do not represent equal Cartesian distances at different radii, and the phase ratio expression also creates a less natural representation for clustering. This explains the much lower polar purity in this implementation.

Part (c): Adaptive K-means versus Fixed Template

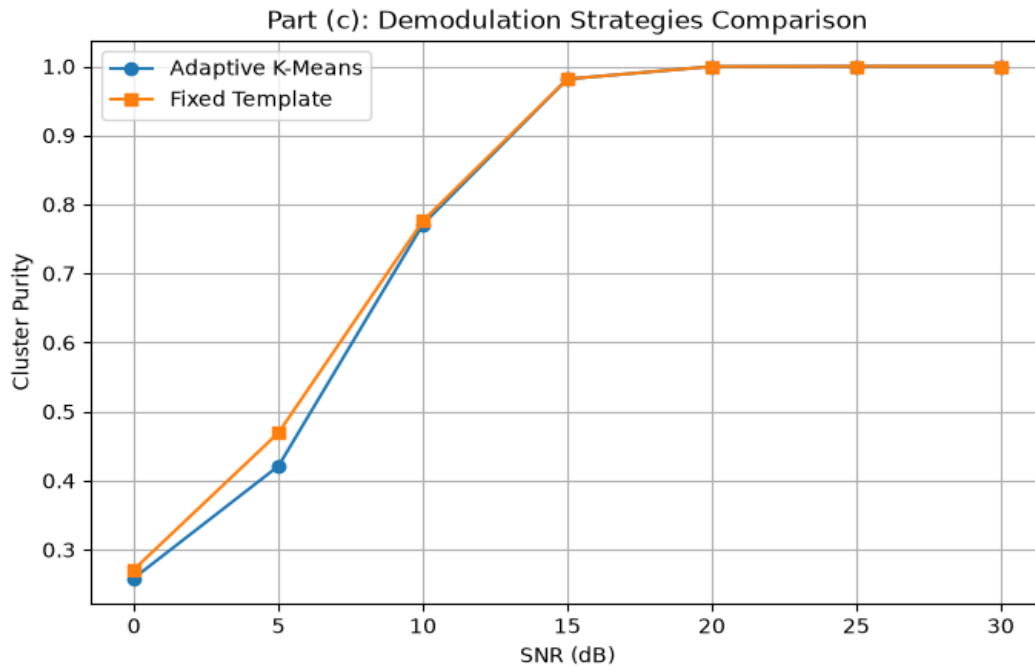


Figure 5 observation. Adaptive K-means fits fresh centroids at every SNR. The fixed-template strategy fits K-means once using the 25 dB Cartesian data and reuses those learned centroids at all SNRs, exactly as required by the assignment. At high SNR, both strategies approach the same purity because received points remain close to the ideal constellation. At low SNR, neighboring symbol clouds overlap, so adaptive centroids can shift or merge toward dense regions. The fixed 25 dB centroids retain the known grid geometry and are slightly more stable for this comparison.

Overall conclusion. Cartesian features are the most natural representation for square 16-QAM. At high SNR, both clustering strategies are reliable; at low SNR, noise overlap limits purity and makes a fixed geometric structure useful. Across both questions, the experiments show that physically meaningful features help, but their reliability depends strongly on SNR.